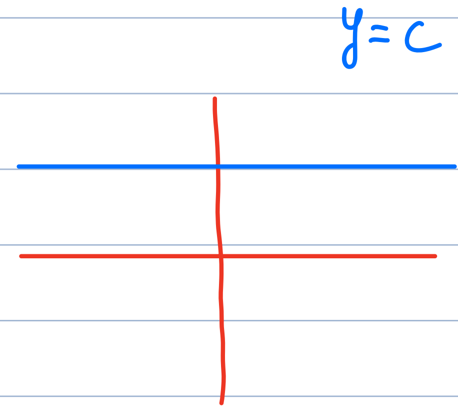
Calc I Lecture 8: first Rules of differentiation

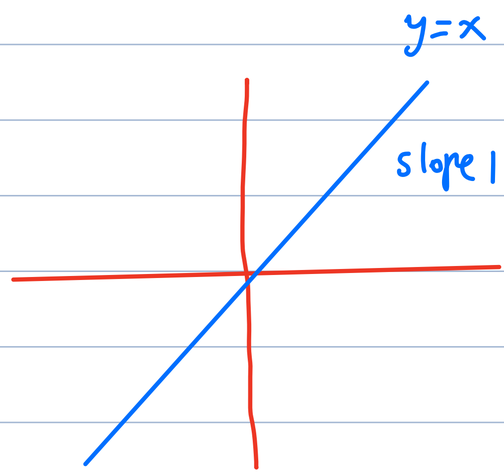
- **Constant** $f(x) = c$ where c is a constant.



$f'(x) = 0$

$$\boxed{\frac{d}{dx}(c) = 0}$$

- $f(x) = x$



$f'(x) = 1$

$$\boxed{\frac{d}{dx}(x) = 1}$$

- $\boxed{\text{Power rule: } \frac{d}{dx}(x^n) = n \cdot x^{n-1}}$

special cases: $n=1$ $\quad \frac{d}{dx}(x^1) = 1 \cdot x^{1-1} = 1 \cdot x^0 = \boxed{1}$

$\frac{d}{dx}(x^1) = \frac{d}{dx}(x) = 1$

$n=2$ $\quad \frac{d}{dx}(x^2) = 2 \cdot x^{2-1} = \boxed{2x}$

$\frac{d}{dx}(x^2) = 2x$

$n=3$ $\quad \frac{d}{dx}(x^3) = 3 \cdot x^{3-1} = \boxed{3x^2}$

in fact, power rule works even if n is not a positive whole number (prove this later)

$$\boxed{\frac{d}{dx}\left(x^r\right) = r \cdot x^{r-1}}$$ for <u>any</u> real number r

Example: Evaluate

a) $\frac{d}{dx}\left(x^{10}\right)$ $r=10$: $\frac{d}{dx}\left(x^{10}\right) = 10 \cdot x^{10-1} = \boxed{10x^9}$

b) $\frac{d}{dx}\left(\frac{1}{\sqrt{x}}\right)$ first rewrite $\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}} \Rightarrow r = -\frac{1}{2}$

$$\frac{d}{dx}\left(\frac{1}{\sqrt{x}}\right) = \frac{d}{dx}\left(x^{-\frac{1}{2}}\right) = -\frac{1}{2}x^{-\frac{1}{2}-1} = \boxed{-\frac{1}{2}x^{-3/2}}$$

c) $\frac{d}{dx}\left(\frac{1}{x^2}\right)$ rewrite $\frac{1}{x^2} = x^{-2} \Rightarrow r = -2$

$$\boxed{\frac{d}{dx}\left(x^r\right) = r \cdot x^{r-1}}$$

$$\frac{d}{dx}\left(\frac{1}{x^2}\right) = \frac{d}{dx}\left(x^{-2}\right) = -2 \cdot x^{-2-1} = -2x^{-3} = \boxed{\frac{-2}{x^3}}$$

d) $\frac{d}{dx}\left(x\sqrt{x}\right)$ rewrite $x\sqrt{x} = x^1 \cdot x^{1/2} = x^{1+\frac{1}{2}} = x^{3/2} \Rightarrow r = \frac{3}{2}$

$$\frac{d}{dx}\left(x\sqrt{x}\right) = \frac{d}{dx}\left(x^{3/2}\right) = \frac{3}{2} \cdot x^{\frac{3}{2}-1} = \frac{3}{2}x^{1/2}$$

$$= \boxed{\frac{3}{2}\sqrt{x}}$$

e) $\frac{d}{dx}\left(\sqrt[3]{x}\right)$ rewrite $x^{1/3}$ $r = \frac{1}{3}$

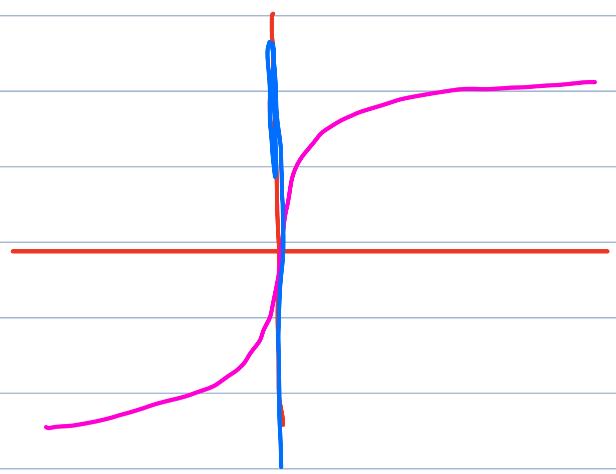
$$\frac{d}{dx}\left(x^{1/3}\right) = \frac{1}{3}x^{1/3-1} = \frac{1}{3}x^{-2/3}$$

$$= \boxed{\frac{1}{3x^{2/3}}}$$

Notice: $\frac{d}{dx}\left(\sqrt[3]{x}\right)$ has a vertical asymptote at $x=0$

In general, if $f'(x)$ has a vertical asymptote at $x=a$

$\Rightarrow$ $f(x)$ has a vertical tangent at $x=a$



$f'(x) \to \infty$

as $x \to 0$

$\Rightarrow$ vertical tangent

more generally, x^r will have a vertical tangent at $x=0$

if $0<r<1$

$$\Rightarrow \frac{d}{dx}\left(x^r\right) = r \cdot x^{r-1}$$

$\leftarrow$ negative

More differentiation rules

- Sum rule: if f is differentiable at a
  g is differentiable at a

  then $(f+g)'(a) = f'(a) + g'(a)$

- Difference rule: if f is differentiable at a
  g is " " "

  then $(f-g)'(a) = f'(a) - g'(a)$

- Constant multiples: if f is differentiable at a,
  and c is any constant, then

  $(cf)'(a) = c \cdot f'(a)$

<u>Example</u> Evaluate $f'(x)$:

a) $f(x) = x^3 - 2x^2 + 5$

b) $f(x) = \dfrac{x^2 - 3}{\sqrt{x}} + x$ (hint: simplify first) solutions:

a) $\frac{d}{dx}(x^3 - 2x^2 + 5)$

sum, difference $= \frac{d}{dx}(x^3) - \frac{d}{dx}(2x^2) + \frac{d}{dx}(5)$

constant multiple $= \underbrace{\frac{d}{dx}(x^3)}_{3x^2} - 2\underbrace{\frac{d}{dx}(x^2)}_{2x} + \underbrace{\frac{d}{dx}(5)}_{0}$

power rule

$= 3x^2 - 2 \cdot 2x = \boxed{3x^2 - 4x}$

b) $f(x) = \frac{x^2 - 3}{\sqrt{x}} + x$ (first, simplify)

$= \frac{x^2}{\sqrt{x}} - \frac{3}{\sqrt{x}} + x = \frac{x^2}{x^{1/2}} - 3 \cdot x^{-1/2} + x$

$x^{3/2}$

$f'(x) = \frac{d}{dx}\left(x^{3/2} - 3x^{-1/2} + x\right)$

sum, difference constant $= \frac{d}{dx}(x^{3/2}) - 3\frac{d}{dx}(x^{-1/2}) + \underbrace{\frac{d}{dx}(x)}_{1}$

$= \frac{3}{2}x^{\frac{3}{2}-1} - 3\left(-\frac{1}{2}x^{-\frac{1}{2}-1}\right) + 1$

$= \boxed{\frac{3}{2}x^{1/2} + \frac{3}{2}x^{-3/2} + 1} = f'(x)$

Example: find $f''(x)$ where $f(x) = \dfrac{x^2 - 3}{\sqrt{x}} + x$

$$f''(x) = \frac{d}{dx}\left(f'(x)\right)$$

$$= \frac{d}{dx}\left(\frac{3}{2}x^{1/2} + \frac{3}{2}x^{-3/2} + 1\right)$$

result from previous problem

$$= \frac{3}{2}\frac{d}{dx}\left(x^{1/2}\right) + \frac{3}{2}\frac{d}{dx}\left(x^{-3/2}\right) + \underbrace{\frac{d}{dx}(1)}_{0}$$

$$= \frac{3}{2}\cdot\frac{1}{2}x^{\frac{1}{2}-1} + \frac{3}{2}\cdot -\frac{3}{2}x^{-\frac{3}{2}-1}$$

$$= \boxed{\frac{3}{4}x^{-\frac{1}{2}} - \frac{9}{4}x^{-\frac{5}{2}}}$$

next in our library of functions...

Derivative of exponential functions

Consider $f(x) = b^x$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx}(b^x) = \lim_{h \to 0} \frac{b^{x+h} - b^x}{h} = \lim_{h \to 0} \frac{b^x \cdot b^h - b^x}{h}$$

$$\frac{d}{dx}(b^x) = \lim_{h \to 0} \frac{b^x(b^h - 1)}{h} = b^x \lim_{h \to 0} \frac{b^h - 1}{h} \quad (*)$$

some number
C

$$\frac{d}{dx}(b^x) = \underbrace{C}_{\text{? mystery number}} \cdot b^x$$

In words: the derivative of an exponential is <u>proportional</u> to the original function

Idea : wouldn't it be nice if $\frac{d}{dx}(b^x) = b^x$ for some b?

In fact, this happens if we choose e as our base.

Definition of e : e is a number such that

$$\boxed{\lim_{h \to 0} \frac{e^h - 1}{h} = 1}$$

Therefore: $\boxed{\frac{d}{dx}(e^x) = e^x}$

$$\frac{d}{dx}(e^x) = e^x \underbrace{\lim_{h \to 0} \frac{e^h - 1}{h}}_{1} = e^x$$

from (*) above

Example : Evaluate the derivative

$$\frac{d}{dx}\left(\frac{x^3 - 2\sqrt{x} - 1}{x^2} + 2e^{x-1}\right)$$

(hint: first simplify)

solution :

First, simplify: $\frac{d}{dx}\left(\frac{x^3}{x^2} - \frac{2\sqrt{x}}{x^2} - \frac{1}{x^2} + 2 \cdot e^x \cdot e^{-1}\right)$

$$= \frac{d}{dx}\left(x - 2 \cdot x^{\frac{1}{2}-2} - x^{-2} + \frac{2}{e} \cdot e^x \right)$$

sum, difference, constant multiple

$$= \frac{d}{dx}(x) - 2\frac{d}{dx}\left(x^{-\frac{3}{2}}\right) - \frac{d}{dx}\left(x^{-2}\right) + \frac{2}{e}\underbrace{\frac{d}{dx}\left(e^x\right)}_{e^x}$$

Now differentiate:

$$= 1 - 2\left(-\frac{3}{2} \cdot x^{-\frac{5}{2}}\right) - \left(-2x^{-3}\right) + \frac{2}{e} \cdot e^x$$

$$= \boxed{1 + 3x^{-5/2} + 2x^{-3} + 2 \cdot e^{x-1}}$$

note: power rule applies when variable is in base (x^n)

not when variable is in exponent (b^x)